SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED
CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 5

Code of Conduct
I BALA S(192524427) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: bala

Assessment Tasks
Short Answer Test
1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly	Uses relevant	Uses vague or	No useful

		relevant cloud examples to strengthen the answer	examples with minor mismatch	partially relevant examples	examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Short answer test

1.Cloud computing:

It is the delivery of computing services such as servers , storage, database , networking ,software and analytics over the internet

Characteristics:

i) ON Demand self service :Users can access computing resources whenever needed without ineraction with the services

ii)Broad network access :cloud services are available over the internet and can be accessed from computers, smartphones, tablets and other devices

iii)Resource Pooling**:** Resources are shared among multiple users using a multi-tenant model, improving efficiency.

iv) Rapid Elasticity**:** Resources can be quickly increased or decreased based on demand

2.

Cloud computing evolved through several technological stages, beginning with improvements in computer hardware and later advancing through internet-based software technologies. These developments laid the foundation for today's scalable, on-demand cloud platforms.

1. **ardware Evolution**

The first stage began with mainframe computers, where many users shared a single powerful computer through terminals. Although efficient, these systems were expensive and centrally managed.

The development of virtualization allowed multiple operating systems and applications to run on one physical server. This significantly improved hardware utilization and became the foundation of cloud infrastructure.

Later, distributed computing and grid computing enabled multiple computers to work together over networks. Instead of relying on a single machine, computational tasks could be shared among many systems, improving reliability and performance.

The expansion of high-speed internet made it possible to access computing resources remotely, removing the need for users to own powerful local hardware.

2. Internet Software Evolution

As the internet matured, software shifted from desktop applications to web-based applications access through browsers.

The introduction of web services and Application Programming Interfaces (APIs) allowed different applications to exchange data and communicate seamlessly over the internet.

Service-Oriented Architecture (SOA) further improved software design by breaking applications into reusable services. This made systems more flexible, scalable, and easier to maintain.

3. Emergence of Modern Cloud Platforms

Combining virtualization, distributed computing, high-speed networking, APIs, and SOA led to the emergence of cloud computing.

Modern cloud platforms provide three primary service models:

- **Infrastructure as a Service (IaaS):** Virtual servers, storage, and networking.
- **Platform as a Service (PaaS):** Development and deployment environments.
- **Software as a Service (SaaS):** Ready-to-use applications delivered over the internet.

Today's cloud platforms also support:

- Automatic resource scaling
- Global data centers
- Pay-as-you-go pricing
- High availability and disaster recovery
- Containers, Kubernetes, serverless computing, artificial intelligence, and DevOps integration.

3.Role of Server Virtualization in Cloud Computing

Server virtualization is the process of dividing a single physical server into multiple Virtual Machines (VMs) using software called a hypervisor. Each VM operates as an independent server with its own operating system, applications, CPU, memory, and storage. Virtualization is one of the key technologies that makes cloud computing possible.

Role of Server Virtualization in Cloud Computing

Server virtualization plays a vital role in cloud computing by enabling efficient, flexible, and scalable use of computing resources.

1. Resource Optimization

- Multiple virtual servers share the same physical hardware.

- Maximizes CPU, memory, and storage utilization.
- Reduces idle hardware resources.

2. Cost Reduction

- Fewer physical servers are required.
- Reduces hardware, electricity, cooling, and maintenance costs.
- Improves return on investment (ROI).

3. Scalability

- Virtual machines can be created, resized, or removed quickly.
- Cloud providers can allocate resources according to user demand.
- Supports elastic scaling during peak workloads.

4. Isolation and Security

- Each VM runs independently.
- Problems in one VM do not affect other VMs.
- Enhances security and system stability.

5. High Availability and Disaster Recovery

- Virtual machines can be migrated between physical servers with minimal downtime.
- Backup and recovery become faster and easier.
- Improves business continuity.

6. Faster Deployment

- New servers can be provisioned within minutes.
- Supports rapid application development and testing.
- Enables DevOps and continuous deployment practices.

Virtualization as an Enabling Technology

Virtualization is the foundation of cloud computing because it:

- Creates multiple virtual servers from one physical server.
- Enables efficient sharing of hardware resources.
- Allows rapid provisioning of virtual machines.
- Supports workload migration and fault tolerance.

- Makes resource pooling possible for cloud providers.

Without virtualization, cloud providers would need dedicated physical hardware for every customer, making cloud services expensive and inefficient.

Cloud Computing as a Service Model

Cloud computing builds on virtualization and adds features that allow users to consume computing resources as services.

These include:

- Infrastructure as a Service (IaaS): Virtual machines, storage, and networking.
- Platform as a Service (PaaS): Development platforms and runtime environments.
- Software as a Service (SaaS): Ready-to-use applications delivered through the internet.

Cloud computing also provides:

- On-demand self-service
- Elastic scalability
- Pay-as-you-go pricing
- Resource pooling
- Broad network access
- Measured service

Relationship Between Virtualization and Cloud Computing

Physical Server – Hypervisor- Virtual machines- Resource pool-Cloud platform-cloud services

4. Comparison of SOAP-Based and REST-Based Web Services

SOAP (Simple Object Access Protocol) and REST (Representational State Transfer) are two common approaches for developing web services. Both enable communication between applications over the internet, but they differ in architecture, communication style, data formats, and their suitability for cloud computing.

SOAP-Based Web Services

SOAP is a protocol that defines strict standards for exchanging structured information between applications. It primarily uses XML for message formatting and can operate over protocols such as HTTP, SMTP, and TCP. SOAP provides built-in support for security, reliability, and transaction management, making it suitable for enterprise applications.

Characteristics

- Protocol-based communication
- Uses XML for data exchange
- Highly secure with WS-Security
- Supports ACID transactions
- Platform and language independent
- More complex and heavier due to XML processing

REST-Based Web Services

REST is an architectural style that uses standard HTTP methods such as GET, POST, PUT, DELETE, and PATCH to access and manipulate resources. REST commonly exchanges data in JSON, although it also supports XML, HTML, and plain text. REST is lightweight, scalable, and widely used for cloud and web applications.

Characteristics

- Architecture-based communication
- Uses HTTP methods
- Supports JSON, XML, HTML, and text
- Lightweight and fast
- Stateless communication
- Easy to develop and integrate

Communication Style

SOAP

- Uses a standardized XML messaging protocol.
- Every request and response follows a predefined XML structure.
- Suitable for applications requiring strict message validation and reliability.

REST

- Uses HTTP as the communication protocol.
- Resources are identified by URLs.
- Operations are performed using HTTP methods such as GET, POST, PUT, DELETE, and PATCH.
- Easier to consume using browsers and web applications.

Data Exchange

SOAP

- XML-only messages.
- Large message size increases bandwidth usage.
- Strict formatting improves consistency but reduces flexibility.

REST

- JSON is the preferred format because it is lightweight and easy to parse.
- Also supports XML, HTML, and plain text.
- Faster data transfer and lower network overhead.

Suitability for Cloud Applications

SOAP is Suitable When:

- High security is required.
- Reliable message delivery is essential.
- Enterprise-level transactions must be supported.
- Applications require strict standards and compliance.

Examples: Banking systems, financial services, enterprise resource planning (ERP), and government applications.

REST is Suitable When:

- High performance and scalability are required.
- Mobile and web applications consume APIs.
- Microservices architecture is used.
- Cloud-native applications require lightweight communication.

Examples: AWS APIs, Google Cloud APIs, Microsoft Azure APIs, social media APIs, e-commerce platforms, and IoT systems.

Advantages and Disadvantages

SOAP

Advantages

- Strong security features (WS-Security)

- Reliable messaging
- Built-in error handling
- Supports distributed transactions

Disadvantages

- Complex implementation
- XML messages are large
- Lower performance
- Higher bandwidth consumption

REST

Advantages

- Lightweight and fast
- Easy to develop and maintain
- Supports multiple data formats
- Highly scalable
- Ideal for cloud and mobile applications

Disadvantages

- No built-in security standard (relies on HTTPS, OAuth, JWT, etc.)
- Less suitable for complex enterprise transactions
- Stateless communication may require additional client-side management

5.

Cloud computing delivers IT resources as services over the internet. These services are categorized into different service models based on the level of control and responsibility provided to users. The four common service models are Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS), and Anything as a Service (XaaS).

i) Infrastructure as a Service (IaaS)

IaaS provides virtualized computing resources such as servers, storage, and networking over the internet. Users rent the infrastructure instead of purchasing physical hardware.

Features

- Virtual servers and storage
- Scalable resources
- Pay-as-you-go pricing
- High flexibility
- Suitable for hosting applications and websites

Real-World Example

Amazon EC2 (AWS) provides virtual machines that allow businesses to deploy applications without owning physical servers.

ii). Platform as a Service (PaaS)

Definition

PaaS provides a complete development platform, including hardware, operating systems, runtime environments, databases, and development tools. Developers can focus on building applications without managing the underlying infrastructure

Features

- Built-in development tools
- Automatic software updates
- Faster application deployment
- Supports collaboration among developers

Real-World Example

Google App Engine enables developers to build, test, and deploy web applications without managing servers.

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iii)Software as a Service (SaaS)

Definition

SaaS delivers ready-to-use software applications over the internet. Users access the software through a web browser without installing or maintaining it.

Features

- No software installation required
- Accessible from any device with internet access
- Automatic updates and maintenance
- Subscription-based pricing

Real-World Example

Microsoft 365 provides online versions of Word, Excel, PowerPoint, Outlook, and other productivity tools through the cloud.

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iv)Anything as a Service (XaaS)

Definition

XaaS (Everything or Anything as a Service) is a broad cloud computing model where almost any IT function is delivered as an on-demand service over the internet. It includes services such as Storage as a Service (STaaS), Database as a Service (DBaaS), Security as a Service (SECaaS), AI as a Service (AIaaS), and many others.

Features

- Flexible service offerings
- Subscription or usage-based pricing
- Easy scalability
- Reduces infrastructure and maintenance costs

Real-World Example

Dropbox provides Storage as a Service (STaaS), allowing users to store, synchronize, and share files securely in the cloud.